

Noisy calibration and latent coverage under log-concavity

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Abstract

Prediction intervals for a latent quantity, such as the true mean of an area that was never sampled, are often calibrated on noisy proxies whose error variances are known. For symmetric unimodal latent errors Anderson’s theorem makes such calibration conservative. We ask what can be guaranteed without symmetry. Over all log-concave latent laws, we characterize the smallest latent radius implied by a noisy coverage level, reduce it exactly to a two-parameter family with closed-form Gaussian convolutions, and certify it numerically. When the noise is small, noisy calibration can undercover the latent target at every coverage level. The radius shortfall is at most $c_q x^{1/2}$ for every noise-to-threshold variance ratio x , and this constant is sharp as $x \rightarrow 0$. A finite-sample slack of 10^{-3} in the calibrated level at $q = 0.9$ removes the shortfall at every noise level, and a slack below 7.9×10^{-4} does not, and above a critical level the noisy threshold may be shrunk. Order-statistic calibration keeps its Beta coverage law under heterogeneous independent noise, by Hoeffding’s comparison of Poisson-binomial and binomial tails. This yields a finite-sample procedure whose conditional reliability holds for every log-concave latent law.

Some key words: Conformal prediction; Deconvolution; Localization; Log-concavity; Measurement error; Small area estimation.

1 Introduction

Suppose a residual W , the latent target minus a centre fitted on independent training data, is observed only through $V = W + e$ with $e \sim N(0, D)$ independent of W and D known. This is the position of a survey statistician who calibrates a prediction interval for the true mean of an unsampled area on the direct estimates of sampled areas (Fay & Herriot, 1979), and of a conformal analyst whose calibration responses carry measurement error (Einbinder et al., 2024; Cohen et al., 2025). The threshold t that gives the noisy scores coverage p is observable. The question is what radius gives the latent residual coverage q .

If W is symmetric and unimodal, Anderson’s theorem (Anderson, 1955) gives $\text{pr}(|W + e| \leq t) \leq \text{pr}(|W| \leq t)$, so the noisy threshold is conservative; Einbinder et al. (2024) use this for conformal prediction with noisy labels. Without symmetry only shape-free statements are available. If the noise has median zero, a noisy interval of level $1 - \alpha$ covers the latent value with probability at least $1 - 2\alpha$, and noisy coverage can exceed latent coverage (Guille-Escuret & Ndiaye, 2024). Cohen et al. (2025) estimate the noise-free threshold by deconvolution, but give no finite-sample guarantee. Worst-case coverage over latent laws that are compatible with noisy observations also underlies the robust empirical Bayes intervals of Armstrong et al. (2022), which constrain moments of the latent law and cover sampled units. Interval methods for small areas likewise target sampled areas (Chatterjee et al., 2008; Yoshimori & Lahiri, 2014; Bersson & Hoff, 2024; Zhang & Tuoto, 2026). We study the problem over the class $\mathcal{L}$ of log-concave laws on the real line, including point masses. This class is closed under convolution, conditioning on intervals, translation and scaling, and it contains the usual models for regression residuals.

After scaling t to 1, write $x = D/t^2$, let $Z \sim N(0, 1)$, and define

$$R_{p,q}(x) = \sup\{Q_q(|W|) : W \in \mathcal{L}, \text{pr}(|W + x^{1/2}Z| \leq 1) \geq p\}, \quad (1)$$

where Q_q denotes the lower q -quantile, and $R_{p,q}(x) = -\infty$ if nothing is feasible, which happens exactly when $\text{pr}(|x^{1/2}Z| \leq 1) < p$. If calibration certifies noisy coverage at least p at threshold t , then $t R_{p,q}(D/t^2)$ is the smallest radius that guarantees latent coverage q for every log-concave latent law. This is sharpness within the class and the single constraint; it is not optimality among all procedures that use the whole calibration sample.

Our results are as follows. The supremum in (1) may be computed over point masses and log-affine densities on a segment, and for these the constraint has a closed form (§2). For every q and every x , $R_{q,q}(x) \leq 1 + c_q x^{1/2}$ with an explicit $c_q > 0$, and the constant is attained as $x \rightarrow 0$. A finite-sample slack $p - q$ of order 10^{-3} makes $R_{p,q}(x) \leq 1$ for all x (§3). A monotone branch and bound certifies $R_{p,q}$ to a relative accuracy of 3×10^{-4} over most of its range. With heterogeneous known noise the order statistic keeps its Beta law, by a comparison of Hoeffding (1956) and Sen (1970), so the construction gives a finite-sample procedure (§4).

2 Reduction to an extremal family

Write $g_x(w) = \text{pr}(|w + x^{1/2}Z| \leq 1) = \Phi\{(1-w)/x^{1/2}\} - \Phi\{(-1-w)/x^{1/2}\}$, so that the constraint in (1) reads $E g_x(W) \geq p$. Let $\mathcal{F}$ be the family of point masses and of laws with density proportional to $e^{\beta w}$ on a segment $[a, b]$.

Proposition 1. $R_{p,q}(x) = \sup\{Q_q(|W|) : W \in \mathcal{F}, E g_x(W) \geq p\}$. *The same holds for any continuous constraint kernel that vanishes at $\pm\infty$, in particular for the average kernel of §4.*

The proof, in the Appendix, conditions W on $[-M, M]$ with M so large that $g_x \leq p$ outside. This keeps the constraint at level p , because it removes only mass where the kernel is below average. It then applies the localization theorem of Fradelizi & Guédon (2004) to the linear, upper semicontinuous functional $\mu \mapsto -\mu(-s, s)$.

By the symmetry $W \mapsto -W$ we may take $\beta \geq 0$. Integration by parts gives, for $\beta > 0$ and $\sigma = x^{1/2}$,

$$\int_a^b e^{\beta w} \Phi\left(\frac{c-w}{\sigma}\right) dw = \frac{1}{\beta} \left[e^{\beta w} \Phi\left(\frac{c-w}{\sigma}\right) \right]_a^b + \frac{e^{\beta c + \beta^2 \sigma^2 / 2}}{\beta} \left\{ \Phi\left(\frac{b-c-\beta\sigma^2}{\sigma}\right) - \Phi\left(\frac{a-c-\beta\sigma^2}{\sigma}\right) \right\}, \quad (2)$$

so $\text{pr}(W + e \leq c)$, and hence the constraint, is a combination of normal distribution functions. Fix the slope and the length and translate W by c . Both $c \mapsto \text{pr}(|W+c+e| \leq 1)$ and $c \mapsto \text{pr}(|W+c| \leq s)$ are log-concave (Prékopa, 1973). So the feasible translations form an interval, and $Q_q(|W+c|)$ is quasi-convex in c and is maximized at one of the interval's two ends. The supremum in (1) is therefore a two-dimensional maximization over slope and length of closed-form quantities, each obtained by two root-findings. This computation and an independent differential-evolution search over the three-parameter family agree to 1.5×10^{-13} at all 73 tabulated values of $x \geq 0.006$.

3 Small noise

The local problem at each endpoint is one-sided. Let

$$c_{p,q} = \sup\{F_Y^{-1}(q) : Y \in \mathcal{L}, \text{pr}(Y + Z \leq 0) \geq p\}, \quad c_q = c_{q,q}.$$

Since $W \mapsto (W - t)/D^{1/2}$ preserves $\mathcal{L}$, the sharp one-sided transfer is exactly $t + D^{1/2}c_{p,q}$ for every D . For the exponential tail $Y_{u,b} = b - E$, with E exponential of rate u and independent of Z ,

$$\kappa_u(b) = \text{pr}(Y_{u,b} + Z \leq 0) = \Phi(-b) + e^{-ub+u^2/2} \Phi(b-u),$$

which is continuous and strictly decreasing in b . Let $b_u(p)$ solve $\kappa_u(b) = p$.

Lemma 1. $c_{p,q} = \sup_{u>0} \{b_u(p) - u^{-1} \log(1/q)\}$. In particular

$$c_q = \sup_{u>0} u^{-1} \log\{e^{u^2/2} \Phi(b_u - u) + e^{ub_u} \Phi(-b_u)\} > 0, \quad b_u = b_u(q).$$

Moreover $q \mapsto c_q$ is nonincreasing.

The proof takes a supergradient u of the concave function $\log F_Y$ at the q -quantile v . Then $F_Y(v + y) \leq \min(qe^{uy}, 1)$, which is the distribution function of an exponential tail with the same quantile and a larger constraint value. Monotonicity in q follows by conditioning away left mass. Ball arithmetic gives $c_{0.80} \in [0.0460730, 0.0460731]$, $c_{0.90} \in [0.0190618, 0.0190619]$, $c_{0.95} \in [0.0084075, 0.0084077]$ and $c_{0.99} \in [0.0013950, 0.0013958]$ (Appendix).

Theorem 1. For every $q \in (0, 1)$ and every $x > 0$, $R_{q,q}(x) \leq 1 + c_q x^{1/2}$. Moreover $R_{q,q}(x) = 1 + c_q x^{1/2} + o(x^{1/2})$ as $x \rightarrow 0$.

Proof of the upper bound. Let $W \in \mathcal{L}$ with $\text{pr}(|W + e| \leq 1) \geq q$, and let $\alpha_R = \text{pr}(W + e > 1)$ and $\alpha_L = \text{pr}(W + e < -1)$, so $\alpha_L + \alpha_R \leq 1 - q$. The law of $Y = (W - 1)/x^{1/2}$ is log-concave with $\text{pr}(Y + Z \leq 0) = 1 - \alpha_R \geq q$. By the definition of c and Lemma 1, $F_Y^{-1}(1 - \alpha_R) \leq c_{1-\alpha_R} \leq c_q$, that is, $\text{pr}(W > 1 + c_q x^{1/2}) \leq \alpha_R$. The same argument applied to $-W$ gives $\text{pr}(W < -1 - c_q x^{1/2}) \leq \alpha_L$. Hence $\text{pr}(|W| \leq 1 + c_q x^{1/2}) \geq q$. $\square$

The lower bound places $W = 1 + x^{1/2}Y$ with Y a near-optimal exponential tail, at a level raised by the exponentially small mass that leaks through the far endpoint (Appendix). The extremal laws have a density that rises exponentially towards a cut-off just outside the threshold, with a boundary layer of width $x^{1/2}$. Noise draws more mass inwards across that endpoint than it pushes out, so the effect is of order $x^{1/2}$ rather than the order x that a Taylor expansion suggests. Symmetry removes it entirely.

The argument splits the noisy failure budget between the two endpoints. It extends to $p \geq q$:

$$R_{p,q}(x) \leq 1 + C_{p,q} x^{1/2}, \quad C_{p,q} = \sup_{\alpha_L + \alpha_R = 1-p} \inf_{\beta_L + \beta_R = 1-q} \max(c_{1-\alpha_L, 1-\beta_L}, c_{1-\alpha_R, 1-\beta_R}),$$

for every x . Numerically the worst split puts the whole budget on one side, so that $C_{p,q} = c_{p,q}$.

Corollary 1. If $C_{p,q} \leq 0$ then $R_{p,q}(x) \leq 1$ for every x . If $c_{p,q} > 0$ then $R_{p,q}(x) > 1$ for all small x , because the proof of the lower bound in Theorem 1 gives $R_{p,q}(x) \geq 1 + c_{p,q} x^{1/2} - o(x^{1/2})$ for every $p \geq q$. The smallest slack $p - q$ with $C_{p,q} \leq 0$ lies in $(4.3 \times 10^{-3}, 5.0 \times 10^{-3}]$, $(7.9 \times 10^{-4}, 1.0 \times 10^{-3}]$ and $(1.5 \times 10^{-4}, 2.0 \times 10^{-4}]$ at $q = 0.80, 0.90, 0.95$, and so does the smallest slack with $R_{q+\delta,q}(x) \leq 1$ for all x ; double-precision root finding puts it at 4.4×10^{-3} , 8.0×10^{-4} and 1.6×10^{-4} .

For split conformal calibration with $K = 110$ and rank $k = 105$, the order statistic certifies $p_k = 0.9036$ with probability 0.96. Then $-0.0572 \leq C_{0.9036, 0.90} \leq -0.057$, so with homogeneous noise the finite-sample procedure never needs widening, and it shrinks the noisy threshold by at least the factor $1 - 0.057x^{1/2}$. The constants $c_{p,q}$ and $C_{p,q}$ are defined for a single Gaussian noise law; for the mixture kernel of §4 the radius is certified directly.

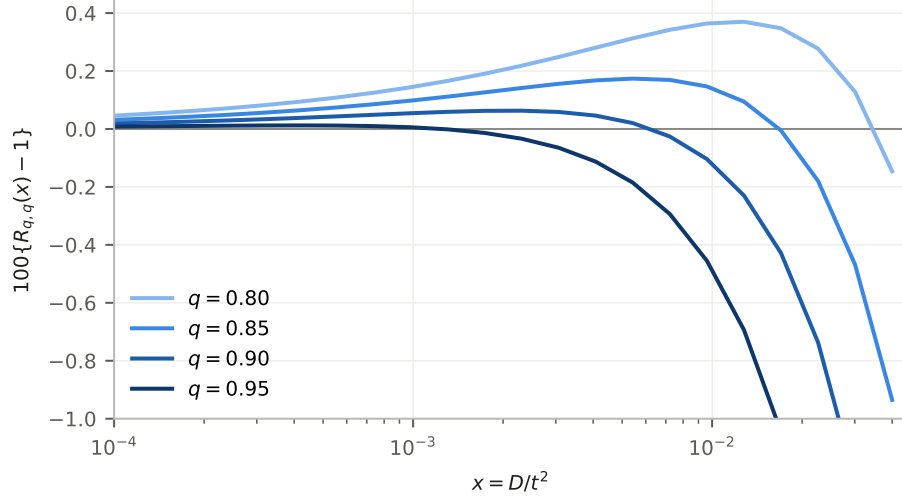


Figure 1: Sharp latent radius relative to the noisy threshold, $100\{R_{q,q}(x) - 1\}$, for four coverage levels. Above zero the noisy threshold must be widened; below zero it may be shrunk.

Figure 1 shows $R_{q,q}(x) - 1$. Grid maximization, which bounds $R_{q,q}$ from below, puts the critical levels at which $R_{q,q} = 1$ at $x^*(q) \approx 0.034, 0.017, 0.0062$ and 0.0012 for $q = 0.80, 0.85, 0.90$ and 0.95 . Below them the noisy threshold must be widened. The largest widening found is 0.37%, 0.17%, 0.063% and 0.012% respectively; the proved bound is $c_q x^{1/2}$. Above them it may be shrunk, substantially once x exceeds about 0.05 (Fig. 2).

4 Certified values and a finite-sample procedure

4.1 Certification

Write a member of $\mathcal{F}$ with $\beta \geq 0$ as $W = b - Y$, where Y has density proportional to $e^{-\beta y}$ on $[0, \ell]$. The ends of the family are included: $\ell = 0$ or $\beta = \infty$ gives a point mass, and $\ell = \infty$ an exponential tail. W is stochastically increasing in β and b and decreasing in ℓ . On a box of parameters, let W_{lo} and W_{hi} be the stochastically smallest and largest corners. The maps $W \mapsto \text{pr}(W + e \leq \pm 1)$ and $W \mapsto \text{pr}(W \leq y)$ are expectations of decreasing functions. Hence, for every law in the box,

$$\begin{aligned} \text{pr}(|W + e| \leq 1) &\leq \text{pr}(W_{\text{lo}} + e \leq 1) - \text{pr}(W_{\text{hi}} + e \leq -1), \\ \text{pr}(|W| \leq s) &\geq \text{pr}(W_{\text{hi}} \leq s) - \text{pr}(W_{\text{lo}} \leq -s). \end{aligned}$$

The density bound $\text{pr}(|W + e| \leq 1) \leq 2 \sup f_W$ is also available. A box whose first bound is below p , or whose second is at least q , contains no law that violates $R_{p,q}(x) \leq s$. Branch and bound over boxes then certifies $R_{p,q}(x) \leq s$. Here b is confined analytically and $\beta, \ell \in [0, \infty]$ are compactified. Between grid points we use the following result.

Proposition 2. *For all $x, h \geq 0$, $R_{p,q}(x + h) \leq R_{p,q}(x) + c_q h^{1/2}$.*

Proof. If W is feasible at noise level $x + h$, then $W + e_h$, with $e_h \sim N(0, h)$ independent, is log-concave and feasible at level x . So $t = Q_q(|W + e_h|) \leq R_{p,q}(x)$. Theorem 1 at scale t then gives $Q_q(|W|) \leq t + c_q h^{1/2}$. $\square$

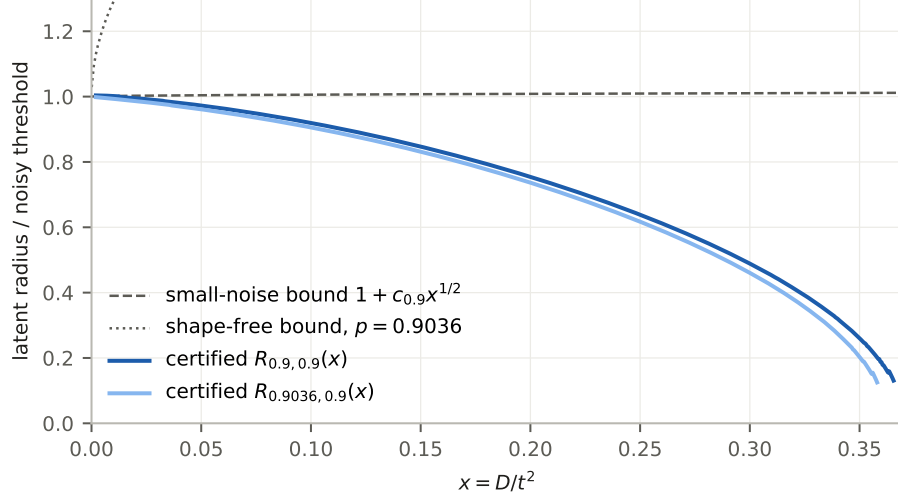


Figure 2: Certified sharp radius $R_{p,0.9}(x)$ for $p = 0.9$ and for the order-statistic level $p = 0.9036$, with the small-noise bound of Theorem 1 and the shape-free bound $1 + x^{1/2}z_{1-(p-q)/2}$, which follows from $|W| \leq |V| + |e|$.

We certified $R_{0.90,0.90}$ and $R_{0.9036,0.90}$ on $x \in [0.002, 0.366]$ with step 5×10^{-4} . The closed forms agree with high-precision quadrature to 10^{-11} , and boxes are cleared with a margin of 10^{-9} , so the values are certified up to floating-point evaluation of the closed forms. For $x \in [0.05, 0.3]$ the certified bound exceeds a feasible value by at most a relative 3×10^{-4} . On $[0.02, 0.3]$ it exceeds the earlier uncertified table by at most 0.15%. The certificate fails only just before the feasibility edges $x \approx 0.370$ ($p = 0.9$) and 0.361 ($p = 0.9036$). Below $x = 0.01$ the lookup also uses Theorem 1 and its $p \geq q$ form, which are within 0.3% of a feasible value there and sharper than the box bounds at $p = 0.9$ (Fig. 2).

4.2 Heterogeneous noise

Let $W_1, \dots, W_K, W_{\text{new}}$ be independent with law $G \in \mathcal{L}$, independent of $e_i \sim N(0, D_i)$, with the D_i fixed and known. The calibration scores $V_i = W_i + e_i$ are then independent but not identically distributed. The following is a case of the comparison of Sen (1970) between order statistics of heterogeneous laws and those of their average; we give the short proof.

Proposition 3. *Let $T = |V|_{(k)}$, let $\bar{F}$ be the average distribution function of the $|V_i|$, assumed continuous, and let p_k be the δ -quantile of the $\text{Beta}(k, K + 1 - k)$ distribution. If $k \geq Kp_k + 1$ then $\text{pr}\{\bar{F}(T) \geq p_k\} \geq 1 - \delta$.*

Proof. $\bar{F}(T) < p_k$ exactly when $N = \#\{i : |V_i| < \bar{F}^{-1}(p_k)\} \geq k$. N is a sum of independent indicators with mean Kp_k , and $k - 1 \geq Kp_k$. By Hoeffding (1956), $\text{pr}(N \leq k - 1)$ is minimized when all success probabilities are equal. Hence $\text{pr}(N \geq k) \leq \text{pr}\{\text{Bin}(K, p_k) \geq k\} = \text{pr}\{\text{Beta}(k, K + 1 - k) \leq p_k\} = \delta$. $\square$

On the event of Proposition 3, the law G satisfies the single linear constraint $\int \bar{g} dG \geq p_k$, with the average kernel $\bar{g}(w) = K^{-1} \sum_i \text{pr}(|w + e_i| \leq T)$. Proposition 1 applies, the box bounds hold term by term, and the heterogeneous rule is

$$s = T R_{p_k - \varepsilon, q}^{\text{mix}}(D_1/T^2, \dots, D_K/T^2).$$

Table 1: Reliability $\text{pr}_{\mathcal{D}}\{\text{pr}(W_{\text{new}} \in C \mid \mathcal{D}) \geq 0.9\}$ with mean width in parentheses; target reliability 0.95

latent law	Fay–Herriot	noisy conformal	log-concave, mean D	heterogeneous
normal	0.960 (3.93)	1.000 (4.96)	0.980 (4.46)	0.980 (4.45)
Laplace	0.927 (3.96)	1.000 (5.15)	0.987 (4.68)	0.987 (4.66)
centred gamma(2)	1.000 (3.95)	1.000 (4.97)	1.000 (4.46)	1.000 (4.45)
truncated Laplace	0.847 (3.89)	1.000 (4.93)	0.973 (4.43)	0.967 (4.42)

Here ε bounds, for each data set, the sup-norm error of the average kernel after compressing the D_i to 32 support points: the maximum over a grid of spacing $h = 10^{-3}$ plus $Mh^2/8$, with M a bound on the second derivative; it was about 2×10^{-4} . The radius R^{mix} is certified for each data set, and the shape-free bound is used if the certificate fails. Replacing the average kernel by the kernel of the mean variance is not conservative in general. At $\bar{x} = 0.2$ with lognormal spread 1 it is 1.4% too short, while the minimum-variance plug-in, which is conservative, is up to 14% too long.

5 Numerical illustration

The target is $\text{pr}_{\mathcal{D}}\{\text{pr}(W_{\text{new}} \in C \mid \mathcal{D}) \geq 0.90\} \geq 0.95$, with $K = 110$ calibration areas, $\text{var}(W) = 1$ and heterogeneous known noise variances $D_i = 0.577 L_i / E(L_i)$, L_i lognormal with log-scale 0.7. Conditional coverage is computed exactly from closed-form latent distribution functions, with 150 replications per law, so the Monte Carlo standard error of a reliability is about 0.018. Table 1 compares four rules:

- the Fay–Herriot tolerance interval with a parametric bootstrap;
- noisy split conformal at rank 105;
- the log-concave rule with the mean variance plugged into the certified $R_{0.9068,0.9}$, which has no guarantee under heterogeneous noise;
- the certified heterogeneous rule.

Every heterogeneous radius was certified, at relative tolerance 10^{-3} in 368 of 600 data sets, 3×10^{-3} in 222 and 10^{-2} in 10, so the shape-free fallback was never used. Certification costs 0.2% of width against the uncertified grid value. Both log-concave rules use the order-statistic level $p_k = 0.9068$ of $\delta = 0.05$. The certified heterogeneous rule meets the target for all four laws. It is only 0.3% shorter than the mean-variance plug-in, so the exact kernel buys validity rather than width here. The order-statistic level accounts for most of the gain over the $p = q$ table: 3.0% of width, of which 1.2% comes from raising p from 0.9036 to 0.9068. The heterogeneous rule is 9.5–10.5% shorter than noisy conformal. It is 12.5–18% longer than the Fay–Herriot tolerance interval, which misses the target under the truncated Laplace law (0.847) and falls short under the Laplace law by about 1.3 standard errors (0.927).

In the finite population **apipop** (Lumley, 2004), 110 of 325 school districts are used for training and 110 for calibration, with 2 or 3 schools sampled per district. Coverage is the share of the remaining 105 districts that are covered. Design variances are estimated with 1–2 degrees of freedom and plugged in as known by every rule, a case that no guarantee covers.

- Reliability over four conditions of 120 replications: the heterogeneous rule 0.942–0.992, the mean-variance rule 0.983–1.000, Fay–Herriot 0.908–0.992 (below target in two conditions).

- The heterogeneous rule at level p_k is 0.8–5.2% shorter than the mean-variance rule with the $p = q$ table, mostly because of the level, and 10–16% longer than Fay–Herriot.

6 Discussion

The results assume that the latent residual is log-concave and that the noise is Gaussian, independent of the latent residual, and of known variance. Log-concavity is not testable from the noisy scores alone, but it is much weaker than the normality used by Fay–Herriot intervals, which missed the target under a truncated Laplace law. Estimated variances require an outer confidence set for the noise kernel; in simulations with χ^2 variance estimates on 4–10 degrees of freedom, plugging them in kept the reliability target for the heterogeneous rule. The small-noise phenomenon of §3 is small in size: its value is that it identifies exactly when noisy calibration is safe without symmetry. The practical gain over noisy conformal comes mainly from the finite-sample slack, which turns the small-noise widening into shrinkage. The exact heterogeneous kernel adds little width, but it is what makes the guarantee hold when the noise variances differ. The constants $c_{p,q}$ and $C_{p,q}$ are certified in ball arithmetic. The branch and bound for $R_{p,q}$ uses double-precision closed forms with a margin, not interval arithmetic.

A Proofs

Proposition 1. Since $\mathcal{F} \subset \mathcal{L}$ it suffices to bound the supremum over $\mathcal{L}$. Let $W \in \mathcal{L}$ satisfy $E g_x(W) \geq p$, and let $s < Q_q(|W|)$, so that $\text{pr}(|W| < s) \leq q - \gamma$ for some $\gamma > 0$. Choose $M \geq s$ with $g_x(w) \leq p$ for $|w| \geq M$, and let W_M be W conditioned on $|W| \leq M$, which is log-concave. With $\varepsilon_M = \text{pr}(|W| > M)$,

$$E g_x(W_M) \geq \frac{p - p\varepsilon_M}{1 - \varepsilon_M} = p, \quad \text{pr}(|W_M| < s) \leq \frac{q - \gamma}{1 - \varepsilon_M} < q$$

for M large. On $K = [-M, M]$ the constraint function $g_x - p$ is continuous, and $\Phi(\mu) = -\mu(-s, s)$ is linear and upper semicontinuous for weak convergence. By the localization theorem (Kannan et al., 1995; Fradelizi & Guédon, 2004), the supremum of Φ over the log-concave laws on K that satisfy the constraint is attained at a point mass or at a log-affine law on a segment. Hence some $F \in \mathcal{F}$ is feasible with $F(-s, s) < q$, that is, $Q_q(|F|) \geq s$. The argument uses only continuity of the kernel and its decay at infinity.

Lemma 1. Exponential tails are log-concave, which gives one inequality. Conversely, let $Y \in \mathcal{L}$ satisfy $\text{pr}(Y + Z \leq 0) \geq p$. If $Y = d$ is a point mass then $d \leq -\Phi^{-1}(p) = \lim_{u \rightarrow \infty} \{b_u(p) - u^{-1} \log(1/q)\}$. Otherwise $F = F_Y$ is continuous and $\log F$ is concave (Prékopa, 1973). Let $v = F^{-1}(q)$ and $G(y) = F(v + y)$.

- The point 0 is interior to $\{G > 0\}$, so the right derivative $u \geq 0$ of $\log G$ at 0 is a supergradient: $G(y) \leq qe^{uy}$ for all y . If $u = 0$ then $G \leq q < 1$, which is impossible, so $u > 0$ and $G \leq G_u = \min(qe^{uy}, 1)$.
- G_u is the distribution function of Y_{u,b_0} with $b_0 = u^{-1} \log(1/q)$. Hence $p \leq E G(-v - Z) \leq E G_u(-v - Z) = \kappa_u(v + b_0)$.
- Since κ_u is decreasing, $v \leq b_u(p) - b_0$.

For monotonicity, let $q' > q$ and let Y be feasible at (q', q') with $v = F_Y^{-1}(q')$. Put $m = (q' - q)/(1 - q)$. Then Y conditioned on $Y \geq F_Y^{-1}(m)$ is log-concave, satisfies the constraint at level $(q' - m)/(1 - m) = q$, and has q -quantile v . Point masses have value $-\Phi^{-1}(q') < -\Phi^{-1}(q) \leq c_q$. Finally, $c_q > 0$ because $u^{-1} \log\{\cdot\} = u/2 + o(u)$ as $u \rightarrow 0$, since $ub_u \rightarrow \log(1/q)$.

Theorem 1, lower bound. Fix $\varepsilon > 0$ and choose u with $b_u(q) - u^{-1} \log(1/q) > c_q - \varepsilon$. For $p' \in [q, (1 + q)/2]$, $b_u(p') \geq b_{\min} > -\infty$. Let $Y = Y_{u, b_u(p')}$ and $W = 1 + x^{1/2}Y$. With $t = 2x^{-1/2}$,

$$\Pr(|W + e| \leq 1) = p' - \Pr(Y + Z < -t) \geq p' - \eta_x, \quad \eta_x = e^{-u(t+b_{\min})/2} + \Phi\{-(t+b_{\min})/2\}.$$

Take $p' = q + \eta_x$. Since $\Pr(|W| \leq s) \leq F_W(s)$, $Q_q(|W|) \geq 1 + x^{1/2}\{b_u(q + \eta_x) - u^{-1} \log(1/q)\}$. As $\eta_x \rightarrow 0$ and b_u is continuous, $\liminf_{x \rightarrow 0}\{R_{q,q}(x) - 1\}/x^{1/2} \geq c_q - \varepsilon$.

The case $p \geq q$. In the proof of Theorem 1, replace $1 - q$ by $1 - p$ for the noisy budget, choose any split $\beta_L + \beta_R = 1 - q$ of the latent budget, and use $F_Y^{-1}(1 - \beta_R) \leq c_{1-\alpha_R, 1-\beta_R}$ on each side. Since $c_{p',q'}$ decreases in p' , the worst case has $\alpha_L + \alpha_R = 1 - p$, and $C_{p,q}$ is nonincreasing in p . The proof of the lower bound applies verbatim with $b_u(q)$ replaced by $b_u(p)$ and $p' = p + \eta_x$, giving $R_{p,q}(x) \geq 1 + c_{p,q}x^{1/2} - o(x^{1/2})$.

Certification. For slopes $\beta' > \beta$ the likelihood ratio $e^{-(\beta' - \beta)y}$ is decreasing, so $Y_{\beta'} \leq Y_\beta$ in likelihood-ratio order. For lengths $\ell < \ell'$, truncation gives $F_\ell \geq F_{\ell'}$. The noisy mass is at most $\sup f_W \int g_x = 2 \sup f_W$, with $\sup f_W = \beta/(1 - e^{-\beta\ell})$. The location b is feasible only in $[-1 + x^{1/2}\Phi^{-1}(p), b_{\max}]$. The lower end holds because $W \leq b$. For the upper end, take $k = 7x^{1/2}$. The density of W increases towards b , so $\Pr(|W| \leq 1 + k) \leq r/(1 + r)$ with $r = (2 + 2k)/(b - 1 - k)$, and $\Pr(|e| > k) = 2\Phi(-7)$.

Ball-arithmetic certificates for $c_{p,q}$ and $C_{p,q}$. Every deciding inequality is evaluated in Arb ball arithmetic; floating point only proposes points.

- $\kappa_u(b)$ decreases in b , and in u because E is stochastically decreasing in its rate. Hence $b_u(p)$ is nonincreasing in u , and on $[u_0, u_1]$, $b_u(p) - u^{-1} \log(1/q) \leq b_{u_0}(p) - u_1^{-1} \log(1/q)$. A ball with $\kappa_u(b) < p$ proves $b_u(p) < b$.
- For $u \geq u_{\max}$ the value is at most $b_{u_{\max}}(p)$. For small u , let $b(u) = \{\log(1/p) + u^2/2 - \log(1 - u^3)\}/u$. Then $e^{-ub+u^2/2} = p(1 - u^3)$, and the Mills ratio gives $\Phi(-b) \leq pu^3$ whenever $\phi\{\log(1/p)/u\} \leq pu^2 \log(1/p)$. This holds on $(0, u_{\min}]$ once it holds at $u_{\min} \leq \log(1/p)/2^{1/2}$, because the left side divided by u^2 increases there. So $b_u(p) \leq b(u)$, and for $p \geq q$ the resulting bound increases in u .
- By Lemma 1, $c_{p',q'} \leq \gamma$ if and only if $1 - q' \geq \varphi_\gamma(1 - p')$, where $\varphi_\gamma(a) = \max[0, \sup_u 1 - \exp\{-u(b_u(1 - a) - \gamma)\}]$. Hence $C_{p,q} \leq \gamma$ if and only if $\varphi_\gamma(a_L) + \varphi_\gamma(a_R) \leq 1 - q$ whenever $a_L + a_R = 1 - p$. Since φ_γ is nondecreasing, it suffices to check this at the ends of an adaptive partition of a_L , with φ_γ bounded above by the same monotone branch and bound in u .

The certificates show $-0.0571845 \leq c_{0.9036, 0.9} \leq C_{0.9036, 0.9} \leq -0.057$, and $C_{q+\delta, q} \leq 0$ for $(q, \delta) = (0.8, 0.005)$, $(0.9, 0.001)$ and $(0.95, 0.0002)$. Lower bounds $c_{q+\delta, q} > 0$ at $\delta = 0.0043, 0.00079, 0.00015$ come from single points u , and $C_{p,q} \geq c_{p,q}$.

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